Product: Open Octave

Team: Team OOP

Abstract

Open Octave is a modular, robotically enhanced educational keyboard system designed to enhance piano learning in schools. The product will consist of octave-sized keyboard modules which can function independently or connect together to form larger keyboard components. Each module will be digitally and robotically augmented to facilitate accessible learning, with light-up keys and a self-pressing mechanism to effectively show beginners how to play. Overall, our system will be composed of three main components: **Hardware & Mechanics** which will be the specialised keys and frame, **Electronics & Firmware** which will be responsible for controlling and connecting hardware components in a meaningful way, and a **Software Application** which teachers can use to remotely configure and control a network of the keyboard modules. Throughout this project, we will need to successfully develop systems for motorised auto-press control, light signal processing, MIDI processing, mode control, module interlocking and connectivity, remote control, multi-PCB communication, and device networking.

1. Pitch

Access to modern music education technology within the public school system has long been hindered by budget constraints (House of Lords Library, 2023). Modern, technologically-enhanced equipment like smart keyboards has struggled to become a permanent fixture of underfunded music departments, impeding the accessibility of learning an instrument within schools.

We want to make smart keyboards completely accessible in the classroom with a flexible, approachable, and budget-friendly educational keyboard tool. Our product, **Open Octave**, is unique in three ways:

1. It leverages robotic actuation (with light-up keys and automatic self-playing) to visually aid students in their learning.
2. It employs a modular design which allows the instrument to grow with the user and allows schools to easily scale their hardware in line with the needs of both students and budget considerations.
3. It comes with an external software application which allows teachers to remotely manage the system of keyboard modules.

All **Open Octave** modules will support three modes:

1. **Piano Mode**, where the keyboard simply functions like a regular piano.
2. **Guided Mode**, where the keyboard lights up while being played to provide sequential fingering cues; the user plays along with the cues
3. **Teaching Mode**, where the keys press themselves down to physically demonstrate how to play a melody and key lights up with different colours indicating which finger should be used to press which key. Wu et al. demonstrate the positive effect of visual cues on learning performance for rhythmic tasks (Wu et al., 2022), further supporting our smart visual assistance proposal.

While each octave-sized keyboard module will be fully functional on its own, students should not be expected to limit themselves to just twelve keys forever. As students' abilities grow, our modular design will allow them to join multiple octaves together to create larger keyboard components - creating a smart keyboard which grows alongside the ability of students. This also ensures that the schools do not incur any unnecessary costs, only upgrading equipment for the students who need it.

Open Octave reimagines the keyboard as a scalable system. Our modular hardware allows schools to start small and grow their inventory as budgets allow, while our robotic enhancements ensure the system is truly accessible to beginner learners - and our accompanying software makes the system manageable within a large classroom setting. With this project, we hope to form part of a sustainable solution for music education in schools.

2. The Team

Ao Zhang is a Computer Science and Artificial Intelligence student with a background in Java and C programming languages. He has got relative experience with LEGO robots and TurtleBots thus willing to work with hardware, but he is also happy to take care of the firmware development part on top of this. He has developed his management skills working as a senior leader of the infPALS scheme.

Laurie Joslin After taking research experience in a startup and academic environment, he has strong experience in

Software Engineering and Data Science. This is coupled with a strong soft skill set in public speaking, team collaboration and project management. Laurie is excited to further his skill in project management by organising roadmaps and sprints but lacks direct hardware experience, albeit is keen to learn more. He is happy to work with firmware and software development.

Ahmad Akasha is an Electronics and Computer Science student with experience in digital system design, analogue circuits, embedded programming, and PCB design. He is comfortable working across the hardware–firmware boundary, with particular strengths in C, Verilog, and microcontroller-based systems. For this project, Ahmad will focus on actuator control, electronics integration, and reliable communication between embedded hardware and higher-level firmware. He has prior engineering internship experience and is confident coordinating technical work across sub-teams during development and testing.

Shuoshuo Wang A computer science student with a background in Python and Java programming languages, he prefers to contribute to the software-related aspects of projects. He hopes to participate in software development tasks related to the overall system functionality and coordinate with the operation of other components within the system. Through the SDP project, he hopes to further accumulate practical software engineering project experience and improve his development and communication skills in a collaborative team environment.

Robin Campbell-Parker Software Engineering student with a proficiency in Java and comfortable with Python and C. His professional experience is grounded in the AI space, but also covers software testing and evaluation. He is excited to work as part of a large team and gain experience with new technologies in a multifaceted project setting, while also utilising his current skills to help out wherever needed.

Ning Liu Computer science student with a good understanding of Java, Python and C programming languages. He is passionate about further developing his technical skills and problem-solving abilities and will assist building the hardware throughout the project.

Alvin Bong Computer Science student with interest in Artificial Intelligence and Software Engineering. He has experience with Data Science and ML projects focused on the finance sector. He is also comfortable with backend development with Java, Spring Boot and Docker and frontend development mainly with React. Outside of technical CS-based knowledge, he has participated in case competitions; he is usually tasked with training models and building dashboards. Alvin is eager to expand his skill sets and contribute to the software side of the project.

Tommy Bardsley Computer Science student with experience mostly in OOP-centred programming languages like Java, C#, Python and a working understanding of Rust. He also has a basic grasp of working with databases, mostly using SQL. Tommy is interested in the security and networking side of Computer science and looks forward to learning new skills and applying his pre-existing skills to develop the firmware in this project.

3. Users

The primary users of our system are beginner piano learners, particularly children and early-stage learners who are being introduced to keyboard instruments and musical theory for the first time. Learning piano at an early stage can often be frustrating, as beginners are expected to translate abstract information such as sheet music or note names into correct physical movements on a keyboard. Our system is designed to support these learners by providing a holistic picture, instantiating sheet music and music theory through visual and physical demonstration.

Consider a beginner learner, Alex, who has just started learning piano at school and is eager to learn as much as possible. Alex does not yet read sheet music and struggles to remember which keys to press when following verbal or visual instructions. With **Open Octave**, Alex's teacher can select a song for him to learn. The system then physically demonstrates the melody in **Teaching Mode** by actuating the keys in sequence whilst simultaneously illuminating them. Alex can watch and listen as the keyboard plays the tune, observing both the order and timing of key presses. After this demonstration, Alex attempts to reproduce the melody in **Guided Mode** using the illuminated keys as guidance. When Alex feels confident enough, he can use **Piano Mode** to practise independently of any robotic assistance. This interaction allows Alex to learn through observation and repetition, closely resembling how a teacher might demonstrate a passage during a lesson.

Several contextual constraints shape the product design for this primary user group. The system must be safe to use around children, as it includes moving mechanical components. Actuation forces are therefore limited, and the system is intended to operate at moderate speeds suitable for beginner-level melodies. Simplicity is also essential: learners should be able to use the system without navigating complex menus or relying on external devices. As such, a lot of the configuration is abstracted away to the teacher via an external application - whilst the core components required for learning are easy to use and understand for the student.

The secondary group of users includes music teachers and educators, who are responsible for administration and setup of the **Open Octave** system. Often-time in music lessons, teachers struggle with being able to help all students learn. This is because students may be playing independently and most likely will have different progression rates. In this case, **Open Octave** provides a interface showing if a student has successfully played a song they have been guided through. This transforms music lessons, with teachers now able to give specialised support to struggling students whilst also providing competent students with another **Open Oc-**

tave module, extending both their knowledge and learning. This facilitation of targeted learning is an important consideration in the design of **Open Octave**.

While the needs of these user groups largely align, there are trade-offs which we have had to consider in our design. Our design prioritises an accessible default interaction for beginners, restricting configuration features and some controls to the network interface accessed by teachers. For a musical education use case, we decided this was the best approach. However it can be acknowledged that an abstraction of control away from individual students could limit students of a higher musical calibre, who perhaps would want to select their own songs to learn to play instead of being limited by the teacher's choice for the class.

Overall, the system is designed to support beginner piano learning by grounding musical instruction in physical action rather than abstract representation. By physically demonstrating melodies and enabling imitation-based learning, the system addresses a genuine educational need that cannot be fulfilled by purely software-based applications. This user-centred perspective provides a strong foundation for deriving detailed user stories and system requirements in later stages of the project.

4. Impact

We envision this system as a proprietary product which could be sold or rented out to educational institutions. We would expect educational institutions to buy (or rent) sets of our modules for their classrooms, allowing them to address the needs of large groups of students trying to learn to play the piano at once. As the needs of these institutions scale, so too would their consumption of the product; for example, when a large proportion of students get good enough to work on multiple connected modules at once, the school might want to purchase more to accommodate this.

This addresses what is currently the biggest barrier to entry for music education within schools: the high cost of quality instruments and educational tools. Research into the current state of music education in the United Kingdom found that the mean yearly budget for music departments in state schools was just £1,865 (Underhill, 2022). This highlights a stark contrast between the offerings of modern technology and the reality of what schools can afford. With even the "budget-friendly" smart keyboards costing several hundred pounds (Welter, 2026), the vision of a classroom where every student has their own instrument is a financial impossibility. Our system bridges this gap by offering a modular, scalable alternative to modern smart keyboards which can be tailored to the financial needs of a school and the abilities of its students.

Another issue we are tackling is the relative inaccessibility of current smart keyboards. A student who has never touched an instrument does not need a full 88-key smart keyboard with all the bells and whistles; they need a simple educational tool which is fitted to their needs and can scale with their ability. This disconnect is especially pertinent for students with intellectual disabilities; Braun et al. highlights the importance of "personalized, adaptable, and customizable technologies" to accommodate those with such disabilities (Braun et al., 2025). We address this divide by incorporating simple visual aids without overcomplicating anything, and offloading complicated controls to the teacher via software.

By creating this new infrastructure for learning the piano within schools, we envision the following wider social implications to be relevant:

- Positive social implications:
  1. **Equal Opportunities:** Our system makes high-quality instruments affordable for every school, helping to level the playing field for students at state schools vs those at independent/public institutions.
  2. **Sustainable Scaling:** Traditional classroom keyboards are often "all-or-nothing" investments that either lack or contain too many features for a classroom setting. Our modular product provides a flexible, long-term asset which encourages departments to commit to piano education with less worries about budget, space constraints, and scalability.
  3. **Inclusive Classroom Participation:** Following the findings of Braun et al., the robotic elements provide the "personalized and adaptable" technology necessary for bridging the gap for students with intellectual disabilities. In a classroom setting, this ensures that these students are not sidelined but can participate alongside their peers in a meaningful way.
- Potential negative social implications:
  1. **Surface-Level Learning:** While visual cues improve immediate performance, research suggests these gains can sometimes disappear once the aids are removed (Emond & Schoppek, 2026). This raises concerns over creating a gamified classroom environment where students focus on reactive light-matching rather than internalising music theory. Over-reliance on simplified visual cues could hinder students' fundamental musical comprehension, leaving them unable to perform independently.
  2. **Visible Talent Hierarchy:** Our modular design could risk creating a clear visual hierarchy, where a student's ability is publicly reflected by the number of octaves they are playing on, where such differences would be less obvious with full-sized keyboards.

To address the concerns outlined above, our design intentionally places teachers at the centre of the learning experience, empowering them to selectively toggle visual aids and robotic assistance based on the educational needs of each student. Also, we would encourage teachers to set up classroom environments with a greater focus on collaboration and inclusive learning, as opposed to more competitive environments - ensuring that students don't end up directly ranked by ability.

5. Outcomes

Our first primary outcome is to ensure our system can provide accurate sound feedback when different keys are being pressed. In musical education, it is crucial for kids to practice on well-tuned pianos so that they can train themselves better on sensing the pitch changes (Liberty Park Music, 2024). As a result, our goal is to develop an accurate sound processing program that outputs sound to headphones and/or a speaker at the correct pitch when a key is being pressed down. One of our main impacts is to make musical and piano learning more accessible to the wider public. Providing accurate sound feedback ensures that more kids have a higher quality learning experience. This will make early music education more accessible, enabling all school children to gain a basic understanding of basic music theory rather than just those whose families can afford a real piano or an 88-key keyboard. To test for this, we will use a digital tuner, which measures how accurate the input sound is compared to the pitch of the actual note. By checking the sound produced by the system when each key is pressed, we can measure how accurate our sound feedback is.

Another primary outcome is the implementation of the two teaching-based modes of operation: **Teaching Mode** and **Guided Mode**. A key requirement of the project is that both these modes reflect the input song (in MIDI format) accurately. The aim is to help kids learn how to play a song using **Open Octave** in an interactive and interesting way. For both of these modes, we will measure our efficacy by manually taking note of the sequence of key presses that's needed to play the song, then when the device starts to press and/or light up the keys in sequence, we can check if the sequence matches our expectations. We also must ensure that these modes effectively convey fingering information via the light colours, meaning each colour will correspond to a specific finger and so when a key lights up in a certain colour it tells the student to use that finger to press the key. There is no need to verify the LED-fingering relation because, for this project, the fingering of the songs will be hard-coded either by the teacher or us as developers. Given the scope of the SDP project, we thought the decision to hard-code LEDs for fingerings makes sense rather than spending resources writing a complex algorithm to do this for us. This algorithm is a tertiary outcome, an extension to our project.

Another primary outcome is the modularity and interconnection functionality. In the Impact section, we stated that to make modern music education technology more accessible, we are going to modularise our product so that more kids can use our device with schools spending less money. Then, when the kids become more familiar with playing the piano or keyboard, their teacher can connect a few of our modules to form a complete keyboard. This is why modularity is key to our product. We can measure this by testing if the interconnection functionality works, so by connecting two **Open Octave** modules and seeing if they function as a whole (one module acting as an octave lower than the other) and if the combined keyboard fulfils the first two primary outcomes.

Our final primary outcome is device networking and a simple software solution. We want to control **Open Octave** modules through a local area network (LAN) in order for teachers to connect to the modules, edit their configurations and choose the modes each module will be on. This will allow for an ideal use case in the classroom. A teacher should be able to easily use a simple GUI application or website that we create that can handle networking to each **Open Octave** module. This would be conducive to successful music lesson planning and execution. We can test this through extensive user testing, where we will have a software application for people to use. If they are happy with this application being functional and intuitive, we will have fulfilled this final primary outcome.

Upon the completion of our primary goals, we will be focusing on our secondary goals. Our first secondary goal is to allow teachers to import new songs as MIDI files, which will allow students to learn songs beyond what is hardcoded into the devices. We will validate this by checking if the three primary goals work for the imported songs. Another secondary goal is to allow teachers to see information about students' playing via the app, allowing them to review and give feedback on what each student is doing from one place.

6. Tasks

Open Octave is decomposed into three subsystems that require different skills and can be worked on in parallel by separate teams throughout the project: the hardware & mechanics, the firmware & electronics, and the software application. Our rough plan for how specific tasks and personnel will be divided between these teams is shown in Figure 1, and the project timeline for each team is shown as a Gantt chart in Figure 2. While these subsystems are designed to be separate and somewhat independent, team dependencies will grow stronger as the project progresses. For this reason, we expect the teams to be flexible, with multiple people ready to move between and work across teams if needed.

At a high level, the hardware team is responsible for the physical manufacturing of the keys and interlocking keyboard frames, the firmware team is responsible for the development of the microcontroller and the overall electronic architecture of the system, and the software team is responsible for developing the application which remotely controls the keyboards.

The ordering of work is planned based on team dependencies. Firmware cannot be meaningfully integrated until keys produce reliable signals. Software cannot be meaningfully built until firmware exposes controllable behaviours such as mode switching and song triggering. The frame must be designed early with modularity in mind, even before networking is implemented at a later stage. This dependency-driven structure explains why, as shown in Figure 2, software work only begins after Demo 1 - with the software team working instead on firmware in the lead up to Demo 1. Firmware team members will move to the software team after Demo 1 and leverage their understanding of the firmware (while also working closely with the remaining firmware team) to define communication protocols and distributed control mechanisms.

Beyond constructing the system, certain team members will also undertake project management and coordination work. Robin will lead overall project coordination and management of the central project GitHub repository, while Ao will keep track of task completion and ensure goals are being met. Ahmad will be in charge of the firmware & electronics team, leveraging his unique understanding of low-level systems to delegate and coordinate work effectively.

User-facing work will also need to be completed; Laurie will take the lead on preparation of demonstrations and presentations for each demo day, while Alvin will lead the poster design and preparation of marketing materials for the industry showcase. Following Demo 2 and the corresponding user feedback, all project team members will ideate together to ensure synchronisation moving forward.

Throughout the project, we expect firmware & electronics development to be the most time-consuming task, although the scope of this work will change over time. In the early weeks, the primary tasks will focus on hardware-firmware development and integration, but as time goes effort will shift to firmware-software development and integration. Overall, we expect roughly 50% of project effort to be concentrated on firmware & electronics, with the remaining work split evenly between hardware and software.

Throughout these teams, tasks, and subsystems, we plan to delegate work according to specific team members' competencies and availabilities. For example, Alvin is primarily competent in software development and will also be periodically away from Edinburgh, so he will start on the software side of things as early as possible to accommodate both his skillset and his remote work requirements.

Task prioritisation is based on our presentation goals for each demo day, with each demo acting as proof of a subset of core system functionalities.

Demo 1 (11/2/2026): Robotic Fundamentals

The goal of Demo 1 is to demonstrate the robotic aspect of **Open Octave**. We aim to have a few keys (at least two) connected to a microcontroller with speaker output, which can demonstrate the following basic robotic functions:

1. Pressing of keys triggering instantaneous audio output.
2. Robotic autopressing and lighting-up of keys; these functions should work in tandem, i.e should happen at the same time.
3. Automated playing of a basic, hardcoded melody using the autopressing and light-up functionality.

Our priority leading up to this demo is to validate and showcase the core mechanical functionality of our system, which will lay the groundwork for all remaining development.

Demo 2 (4/3/2026): Full Octave and Operating Modes

The goal of Demo 2 is to demonstrate a framed 12-key octave which is able to showcase all three modes of operation. That is, the keyboard should demonstrate the following functionalities:

1. **Piano Mode:** The keyboard should be fully functional as a basic keyboard, where a user can press keys and play music without any robotic assistance.
2. **Guided Mode:** The keyboard should be able make keys light up in a specific sequence based on one or two preset hardcoded songs; the keyboard should light up one key at a time and wait until each key has been pressed before lighting up the next one; the light colours should correctly correspond to which fingers should be used to play each note.
3. **Teaching Mode:** The keyboard should be able to automatically play a preset, hardcoded song in real time, utilising the auto-press and light-up functions to demonstrate how to play this song to the user.
4. The keyboard should be able to communicate with a computer via USB to switch between modes and activate functions.

Since Demo 2 is for user-testing and feedback, we have designed the Minimum Viable Product for this demo based on the feedback we want to obtain from users. We primarily want to ascertain whether the educational aspects of our system are actually helpful to complete beginners trying to learn how to play the keyboard.

Demo 3 (25/3/2026): Modularity and Networking

The goal of Demo 3 is to demonstrate the full vision of **Open Octave**: multiple keyboard modules physically connected and digitally networked to create a scalable infrastructure for learning. We aim to demonstrate the following:

1. Have two independently functional keyboard modules.
2. Have the keyboard modules physically interlock to form a fully-functional two-octave keyboard.
3. Have the larger two-octave keyboard showcase more complex songs than what is already possible on a single octave.

4. Have an external software application which can be used to remotely control the keyboard modules (on/off, volume, etc.), activate functions, configure modes, see module connection information, and maybe upload and select songs as MIDI files, and get feedback based on the students' output.

Our goal for Demo 3 (Industry Day) is, of course, to have a fully functional system which clearly demonstrates the utility and novelty of our product. If possible, we hope to get feedback from real music educators and try to ascertain the real-world viability of our system.

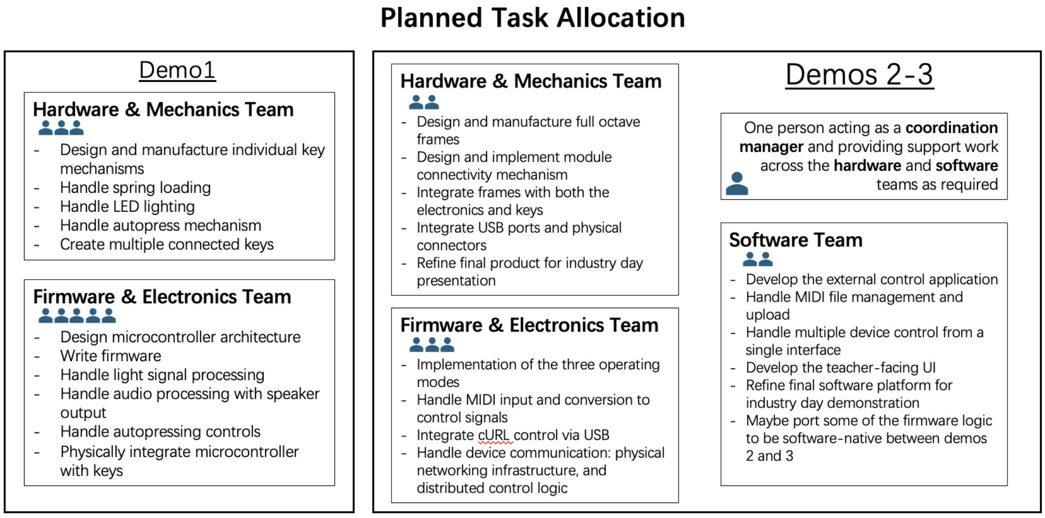

Figure 1. Task allocation across teams and subsystems

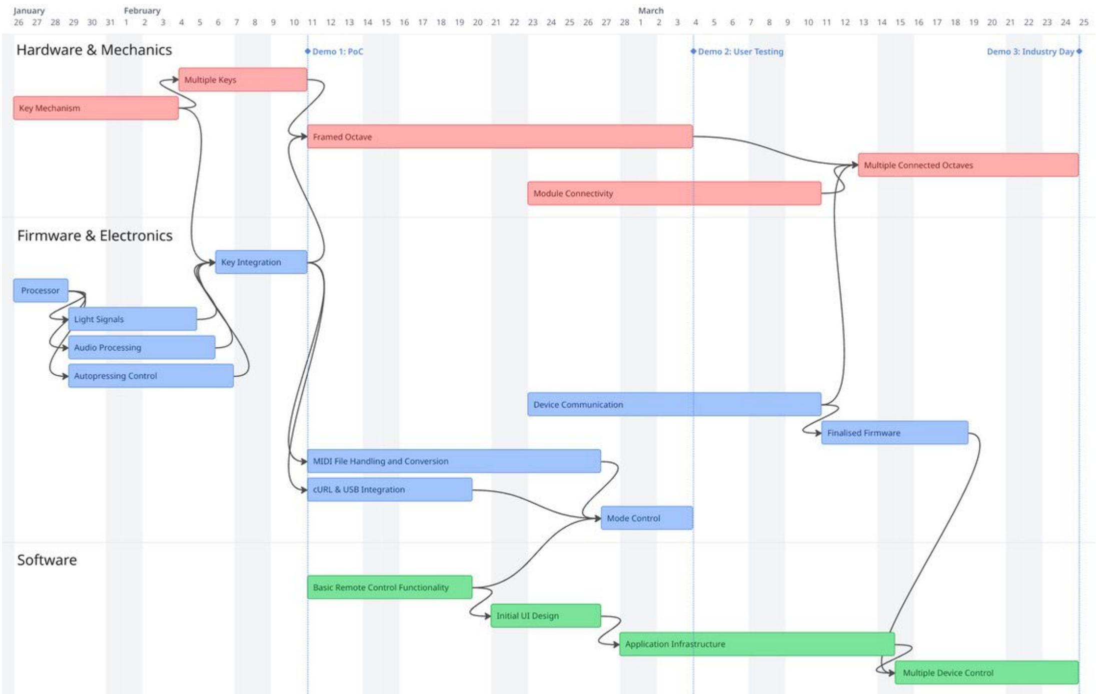

Figure 2. Gantt chart showing high-level task ordering and dependencies

7. Risks

Risk	Likelihood	Impact	Consequences	Contingency Plan
Electrical faults (short circuit, overheating, cable damage)	Low	High	Posing serious safety hazards to users and the environment, and may lead to product failure	We will use a low-voltage power supply and include protective circuitry such as fuses, current limiting, and thermal protection where appropriate. We will improve PCB layout and spacing to reduce short-circuit risk, and ensure wiring insulation, strain relief, and fixation to prevent cable damage.
Keys are too stiff or uncomfortable to press	Medium	Medium	Excessive key force causes fatigue or discomfort, affecting the learning experience and product usability.	We will optimise the key feel by conducting pressure and stroke tests; we will adjust the hand feel design based on user test feedback.
Key caps or enclosure cracking (3D printed parts)	Medium	Medium	It may lead to structural failure and may produce sharp edges, posing a risk of scratches to users	To mitigate this, we will avoid thin walls and sharp corners in the 3D print design, and add basic fillets/chamfers in CAD. We will sand any sharp edges
If wireless connections are used, latency could cause lights and sounds to be out of sync.	Medium	High	Asynchronous lighting and sound can lead to a chaotic learning pace, reduce interactivity and teaching effectiveness, and affect the overall learning experience	We will test the prototype in typical demo conditions and measure whether sync issues are noticeable. If latency is problematic, we will simplify the system: reducing wireless features, using local timing on the device, or switching to a wired connection for the demo. We will then define an acceptable tolerance for the assignment
LED brightness or flicker causing visual discomfort	Low	Medium	Eye strain, discomfort for sensitive users.	We will set a moderate default brightness. We will avoid very low PWM (Pulse-Width Modulation) settings and confirm visually that flicker is not obvious in normal use. We will add a caution note for sensitive users and limit demo duration if needed.
Loose or incorrect modular connections between keyboard units	Medium	High	Modules may detach; signal loss; electrical connection issues.	Keyed connectors; magnetic alignment; mechanical latching; connection detection in firmware.

Table 1. Summary of Risks, Consequences, and Contingency Plans

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